SKIN DISEASE CLASSIFICATION – PROJECT REPORT

Done by Sivaharini K

ABSTRACT

Skin diseases are among the most common health conditions affecting people worldwide. This project presents a Skin Disease Classification System using Deep Learning. The system utilizes a Convolutional Neural Network (CNN) trained on the HAM10000 dataset containing 10,015 dermoscopic images belonging to seven different skin disease categories.

The developed application allows users to upload skin lesion images and obtain instant disease predictions along with confidence scores and disease-related information. The system is deployed using Streamlit, providing a simple and interactive web interface. The project demonstrates the practical application of Deep Learning in healthcare and highlights the potential of AI-assisted diagnosis systems.

INTRODUCTION

Artificial Intelligence (AI) has become one of the most influential technologies in modern healthcare. Deep Learning, a subset of AI, has achieved significant success in image recognition and medical diagnosis applications. Skin diseases affect millions of individuals globally, and early identification is critical for effective treatment.

The Skin Disease Classification System aims to assist in the preliminary identification of skin diseases using dermoscopic images. By leveraging Convolutional Neural Networks, the system can automatically learn important visual features from images and classify them into predefined disease categories. This project provides a user-friendly web application that enables quick and efficient disease prediction.

TOOLS AND TECHNOLOGIES USED

- **Python – Used as the main programming language for developing the project.**
- **TensorFlow – Used for building and training the deep learning model.**

- **Keras – Used to create and manage the Convolutional Neural Network (CNN).**
- **NumPy – Used for numerical operations and image data processing.**
- **Pandas – Used for reading and handling the dataset.**
- **OpenCV – Used for image preprocessing, resizing, and color conversion.**
- **Scikit-learn – Used for data splitting, label encoding, and model evaluation.**
- **Streamlit – Used to develop the interactive web application.**
- **Visual Studio Code (VS Code) – Used as the development environment.**
- **HAM10000 Dataset – Used for training and testing the skin disease classification model.**
- **Convolutional Neural Network (CNN) – Used for classifying skin diseases from dermoscopic images.**

DATASET DESCRIPTION

The project uses the HAM10000 (Human Against Machine with 10000 Training Images) dataset, which is a publicly available benchmark dataset for skin lesion analysis.

Dataset Details:

- **Dataset Name: HAM10000**
- **Total Images: 10,015**
- **Number of Classes: 7**
- **Image Size: 96 × 96 × 3**

Disease Categories :

1. AKIEC – Actinic Keratoses and Intraepithelial Carcinoma

2. BCC – Basal Cell Carcinoma

3. BKL – Benign Keratosis-like Lesions

4. DF – Dermatofibroma

5. MEL – Melanoma

6. NV – Melanocytic Nevi

7. VASC – Vascular Lesions

The dataset contains high-quality dermoscopic images that are widely used for skin disease classification research and development.

METHODOLOGY

- **Data Collection**
  HAM10000 dataset is collected and prepared for training.
- **Image Preprocessing**
  Images are resized and normalized before training.
- **Model Training**
  A CNN model is trained using TensorFlow and Keras.
- **Prediction**
  The trained model predicts the skin disease from the uploaded image.
- **Deployment**
  The model is integrated into a Streamlit web application for user interaction.

This methodology enables efficient skin disease classification using deep learning techniques.

IMPLEMENTATION

The Skin Disease Classification System is implemented using Python and Deep Learning techniques. The HAM10000 dataset is used to train a Convolutional Neural Network (CNN) model. Images are preprocessed and fed into the model for learning disease patterns.

The trained model is integrated into a Streamlit web application. Users can upload a skin lesion image through the interface, and the system predicts the disease category along with confidence scores and relevant information. The application provides a simple and user-friendly platform for skin disease classification.

INSTALLATION OF REQUIRED LIBRARIES

The required Python libraries were installed using the pip package manager. These libraries include TensorFlow, Keras, NumPy, Pandas, OpenCV, Scikit-learn, and Streamlit. They were used for deep learning, image processing, data handling, model evaluation, and web application development.

Example Installation Command:

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pip install tensorflow keras numpy pandas opencv-python pillow scikit-learn streamlit

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OUTPUT

The developed Skin Disease Classification System successfully predicts skin diseases from uploaded dermoscopic images. The user uploads an image through the Streamlit interface, and the trained CNN model analyzes the image to generate predictions.

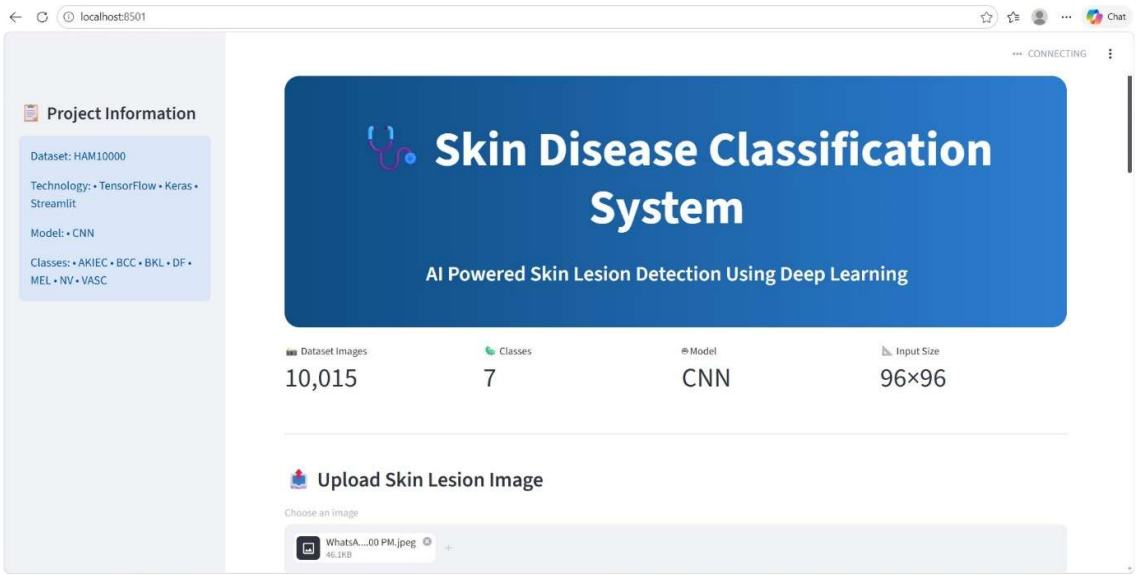


Figure 1: Home Page of the Application

The system displays:

- **Predicted Disease Name**
- **Confidence Score**
- **Risk Assessment**
- **Disease Information**
- **Top Prediction Results**

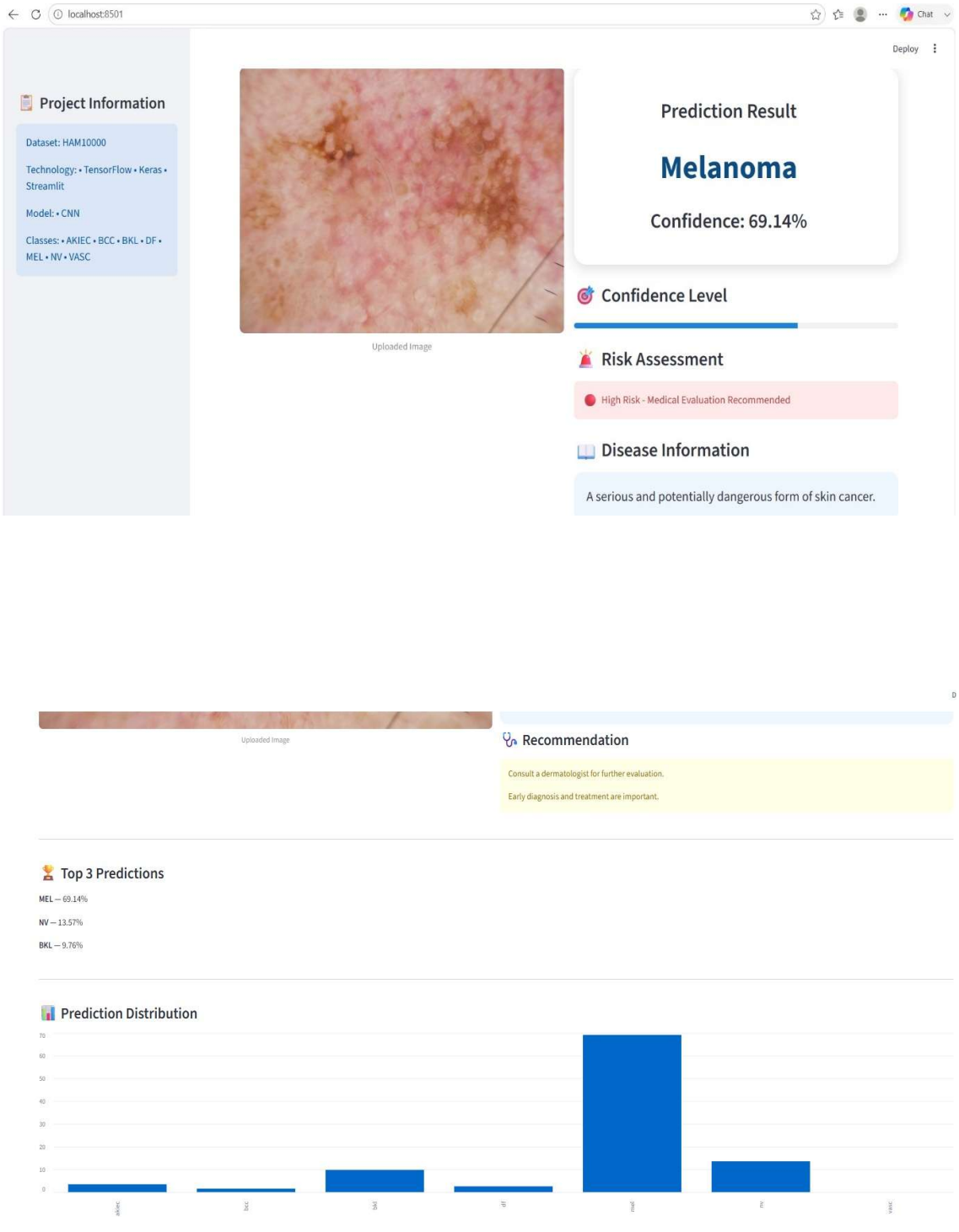
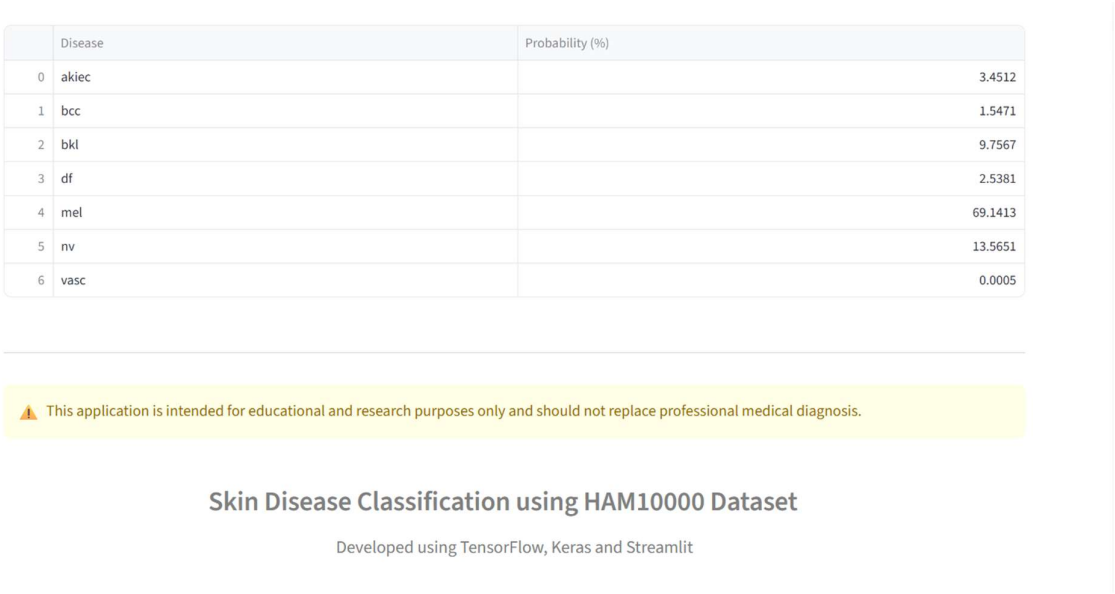


Figure : Image Upload Interface & Disease Prediction

The output is generated instantly, providing users with a simple and interactive experience for skin disease classification.



	Disease	Probability (%)
0	akiec	3.4512
1	bcc	1.5471
2	bkl	9.7567
3	df	2.5381
4	mel	69.1413
5	nv	13.5651
6	vasc	0.0005

TROUBLESHOOTING TABLE

Problem	Cause	Solution
Image not uploaded	Unsupported file format	Upload JPG, JPEG, or PNG images
Model not loading	Missing model file	Ensure skin_disease_model.h5 is available
Incorrect prediction	Poor image quality	Upload a clear skin lesion image
Application not starting	Missing libraries	Install all required Python libraries
Streamlit error	Configuration issue	Restart the application

SCOPE OF THE PROJECT

The scope of this project is to develop an AI-based system for skin disease classification using deep learning techniques. The system helps users identify possible skin diseases from dermoscopic images and provides quick predictions through a web application. It can be used for educational purposes, research, and preliminary screening. In the future, the system

can be enhanced by adding more disease categories, improving accuracy, and integrating mobile or cloud-based deployment.

CONCLUSION

The Skin Disease Classification System using Deep Learning successfully classifies skin diseases from dermoscopic images using a Convolutional Neural Network (CNN). The model is trained on the HAM10000 dataset and integrated into a Streamlit web application for easy user interaction. The system provides disease predictions along with confidence scores and relevant information. This project demonstrates the effectiveness of deep learning in medical image analysis and highlights the potential of Artificial Intelligence in supporting healthcare applications. While it is not a replacement for professional medical diagnosis, it serves as a useful tool for preliminary skin disease screening and awareness.

REFERENCES

- **HAM10000 Dataset – Kaggle.**
- **TensorFlow Documentation.**
- **Keras Documentation.**
- **Streamlit Documentation.**
- **OpenCV Documentation**